Jul 24, 2026

llm-d & vLLM code review and Innovation work - Transcript

00:00:19

Alberto Perdomo: Hermes doing fine.

Ramesh Doddaiah: Hello Alberto. How are you?

Alberto Perdomo: How are

Ramesh Doddaiah: Good, good, good.

Alberto Perdomo: you?

Ramesh Doddaiah: I just uh morning I went to workouts and came. I'm very hungry already.

Alberto Perdomo: Okay, let me uh Did you see what I just sent you?

Ramesh Doddaiah: I just logged in and I'm just seeing the chart. Uh that's unbelievable. This a purple, right?

Alberto Perdomo: Yeah. Yeah. Yeah.

Ramesh Doddaiah: uh untuned and tuned. That's the only

Alberto Perdomo: Uh yes. I mean there is quite a a big difference.

Ramesh Doddaiah: difference.

Alberto Perdomo: Um it is uh I change I mean now Seph is going through uh I'm not sure if it's RDMA but it's the the fast nicks that are present in the

Ramesh Doddaiah: Okay. Okay.

Alberto Perdomo: cluster.

Ramesh Doddaiah: But this is definitely good. This is the amazing story that we want to show in the uh um uh product management. So it matches our hypothesis.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: We are in the right direction.

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Ramesh Doddaiah: So we can fine-tune more u maybe the expert we can talk to boss or other people who are the expert and squeeze a bit more out of

Alberto Perdomo: Yeah. Yeah.

Ramesh Doddaiah: that.

Alberto Perdomo: But it's I mean it's good to see that uh um that basically the seph performs in the same way as a local MVME because it's I mean it's distributed so it's always good.

Ramesh Doddaiah: Yes.

Alberto Perdomo: This has no replication by the way. I removed the replication for now to this initial testing in the SE. There is three nodes but only one replica.

Ramesh Doddaiah: Uhhuh.

Alberto Perdomo: There is no no three replicas.

Ramesh Doddaiah: That's perfectly fine.

Alberto Perdomo: So yeah that that that was taking a lot of the bandwidth.

Ramesh Doddaiah: That's perfectly fine. Okay.

Alberto Perdomo: So

Ramesh Doddaiah: But how much how much is the bandwidth you are getting? Earlier we used to get 300

Alberto Perdomo: uh no I need to I need to rerun the test to measure that because for some reason the the service

Ramesh Doddaiah: Mbps.

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Alberto Perdomo: account that measures the se uh metrics uh had had some uh arbback uh which is the cluster monitoring authorization thing. It has some lack of permissions. So I need to fix that in order to measure the the se. Um but um in my initial runs uh I I asked Claude to do a a small uh stress test and it around uh 13 Gbps something like

Ramesh Doddaiah: Mhm.

Alberto Perdomo: this gigabits.

Ramesh Doddaiah: Gigabytes. gigabits. Okay. Okay. Okay. But still uh what we used to see earlier with the 300 and what we're getting with 16 Gbps,

Alberto Perdomo: Yeah.

Ramesh Doddaiah: it's Apple

Alberto Perdomo: Yeah.

Ramesh Doddaiah: oranges.

Alberto Perdomo: It was around uh 1.3 gigabytes.

Ramesh Doddaiah: Okay.

Alberto Perdomo: So it was almost a little bit over four times

Ramesh Doddaiah: Okay.

Alberto Perdomo: better.

Ramesh Doddaiah: Okay. But what about the latency? So if you tell me the latency, I can tell you if it's RDMA is there or not.

Alberto Perdomo: Uh let me let me share my screen real quick.

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Alberto Perdomo: Uh share screen.

Ramesh Doddaiah: Okay.

Alberto Perdomo: Share entire screen. Let's wait a bit because it will be blurry at first. Yeah. So uh it's here.

Ramesh Doddaiah: Okay.

Alberto Perdomo: Uh for latencies it's again uh the TTFD is the same as as uh well this is end to end P95 but TTFTP95 is the same as uh as uh CPU plus MVME.

Ramesh Doddaiah: Okay. Okay. The green and purple are same.

Alberto Perdomo: Yeah, exactly are basically the same and uh uh oh this yeah a lot

Ramesh Doddaiah: Okay.

Alberto Perdomo: of external KV transfer. So, and compared to to CPU, which is the this line, we can tell that this bit here, it's probably due to the the in this case MVME, in this case, SE, but it's it's pretty cool to see like it's on par here with let me this,

Ramesh Doddaiah: Yes.

Alberto Perdomo: this, and this. It's basically the same thing as an

Ramesh Doddaiah: Okay.

Alberto Perdomo: MVME.

Ramesh Doddaiah: How many pods? How many uh PD do we have?

Alberto Perdomo: Oh, this this is the regular that I did just one replica TP2.

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Ramesh Doddaiah: Single

Alberto Perdomo: So, yeah, we need to do uh multiple um multiple replicas. But in this case, I'm rerunning the whole matrix uh with all of these uh with the uh tune. So, we get uh like uh we will generate a report like this one but with good results for for each of the models. And with that, I think we could um we could probably close the CPU offloading

Ramesh Doddaiah: Yes, absolutely. I agree with that.

Alberto Perdomo: ticket.

Ramesh Doddaiah: But it's just that I'm not uh sure whether um this is the best SE performance that we can get. As a result, if uh if we tune a SE later next week or so, then we may have to rerun the SE related um test cases,

Alberto Perdomo: Yeah. Yeah. Yeah.

Ramesh Doddaiah: right? Okay.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: But um we are in the right direction that I agree there's no doubt about that. It's just that we don't know how much best we can get as of now.

Alberto Perdomo: Yeah, exactly.

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Ramesh Doddaiah: Okay.

Alberto Perdomo: But it's good because with Vun we talked the other day that that he was not able to to see any any SE like any good performance in SE. Uh so at least we are we are seeing right now performance parity across SE and MVME.

Ramesh Doddaiah: Yes.

Alberto Perdomo: So that's good.

Ramesh Doddaiah: Yeah. So we just have to figure out if you have more P and U prefill and decode. So the prefill is destaging to the um um se on one node and the decode is pulling the same data from a different node but the same se file system. So that experiment is a right uh experiment to stress u safe

Alberto Perdomo: Yeah. Yeah. Definitely. Definitely. Yeah. We will have to I think what we need to do is uh um finish the single replica to have

Ramesh Doddaiah: features.

Alberto Perdomo: baselines and see how things scale. Uh do multi-replica and once we have multi-replica but single node and aggregate, we can move to to this aggregated PD.

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Alberto Perdomo: uh uh to see this kind of benefits which I think that it will it will show

Ramesh Doddaiah: Yeah.

Alberto Perdomo: better.

Ramesh Doddaiah: So when you say disagregated is across nodes is it?

Alberto Perdomo: Uh disagregated is basically when you have different pods for preill and

Ramesh Doddaiah: Yeah.

Alberto Perdomo: decode.

Ramesh Doddaiah: But uh within the node or across node which one are you giving more weightage?

Alberto Perdomo: Uh I mean both are disagregated but in our case we can test uh different combinations. We can single node and multiode.

Ramesh Doddaiah: Because if it is a multi node, no, we have to go through the rock key. Yeah.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: So that will add another latency or variable to that. So that we need to also looks like um um most probably you'll be able to complete

Alberto Perdomo: Yeah.

Ramesh Doddaiah: the non seated testing benchmarking soon.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: But the SE uh once we come up with the PD um within a node across node and rock key without rock key and TCP so that is its own different matrix altogether.

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Alberto Perdomo: Yeah, definitely.

Ramesh Doddaiah: Okay. Okay. Cool. Yeah.

Alberto Perdomo: Definitely.

Ramesh Doddaiah: So I think we're on the same page. I I want to use this meeting for our personal work. Okay. So let's wind it down in 2 minutes so that at least we have next 50 minutes for our

Alberto Perdomo: Yeah.

Ramesh Doddaiah: uh uh cool work. So this one apart from this um uh yeah um Ashish was saying Bo is coming uh this Sunday. Uh so he said if you have any questions you can definitely ask him he said so uh he is

Alberto Perdomo: Yeah.

Ramesh Doddaiah: our uh he said he's our local expert on the se but we have not talked to any other

Alberto Perdomo: Yeah. Yeah.

Ramesh Doddaiah: guys the safe so we don't know um much about that um what do you call uh all the mesh part of the se because there's no valid guide for the se

Alberto Perdomo: Yeah.

Ramesh Doddaiah: that's a problem we shall we write a valid guide for se

Alberto Perdomo: Yeah.

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Alberto Perdomo: Yeah.

Ramesh Doddaiah: based on our experience Yes,

Alberto Perdomo: That could that could be that could be nice actually. Um but yeah I think was

Ramesh Doddaiah: we we learned more.

Alberto Perdomo: will help us a lot because he knows a lot about networking and fast networking.

Ramesh Doddaiah: Yeah.

Alberto Perdomo: So yeah for distributed file system it will be um uh very much uh appre appreciate it.

Ramesh Doddaiah: Okay, cool.

Alberto Perdomo: So

Ramesh Doddaiah: We can also include him and publish this uh document. This is the common problem I guess future.

Alberto Perdomo: yeah.

Ramesh Doddaiah: Okay,

Alberto Perdomo: Yeah.

Ramesh Doddaiah: so we're on time and uh uh now coming to our work cool work.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: What exactly you wanted to focus

Alberto Perdomo: Um there are a lot of things that can be done.

Ramesh Doddaiah: on?

Alberto Perdomo: Um but uh yeah, I think we need to start first discussing what kind of projects we want to tackle with this.

Ramesh Doddaiah: Okay.

Alberto Perdomo: uh because 100% sure of all the opportunities that there we might have. Um but let me check because you shared a few things.

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Ramesh Doddaiah: uh how about that you will take a step back and see what the VLM

Alberto Perdomo: Uh

Ramesh Doddaiah: or LLMD uh gives more priority. So even though we might like to do some part of the work but if a VLM and LMD thinks this is most important future feature for the next 6 months or one year and if you put our bandwidth into into those things so that way we'll be able to work on more interesting projects that is lot of people would have thought of so we can do both definitely we can go in our direction for me I definitely will put more um not the more I

Alberto Perdomo: Yeah.

Ramesh Doddaiah: will definitely will put my time to understand the complete KV cache offloading stack that I'm more interested to optimize each and every layer in this KV cache offloading either it is moon cake or LM cache or the uh VLM or something new if you want to do definitely I will look into that in that direction um um some of the patents which I have done I have done hundreds of KV cash patents so it'll come up maybe in one or two years it'll show up in all the Google Scholar profile.

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Ramesh Doddaiah: So um activity based activity based KV cash offloading I feel it's one of the core idea uh that is why I said that's one of the problem which I said we can work on that among the two I think the that has more weightage I will definitely put into more thoughts and start doing some of the coding and share with you all the things on that on that direction if you want we can also go deeper into that problem I can explain you what exactly is the problem um uh all those things or if you want to bring up anything more interesting. Um, feel free to bring up now or later

Alberto Perdomo: Okay, cool. Uh, let me uh Okay, this is transcribed and let me uh enable recording of

Ramesh Doddaiah: also

Alberto Perdomo: this. So, we record this. Okay, cool. Awesome. So, we have recorded. Yeah. So I think that the that the activity based uh KB cache offloading uh is definitely something very interesting. So I think we can we can start working off that.

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Alberto Perdomo: And um sometimes with these kind of projects, I feel like uh it doesn't really matter exactly where you start with uh

Ramesh Doddaiah: Mhm.

Alberto Perdomo: eventually find the more interesting work along the along the way.

Ramesh Doddaiah: Okay. Okay. That is fine because um uh the way my experience is

Alberto Perdomo: So

Ramesh Doddaiah: um uh the product which I was working on was used by was used by each and every stock market across the world.

Alberto Perdomo: Okay.

Ramesh Doddaiah: So any stock market all the data comes to that product.

Alberto Perdomo: Wow.

Ramesh Doddaiah: And I I had developed all this scheduleuler for that with thousands of CPUs and all these data storage and um there what happens is if I talked to

Alberto Perdomo: Wow.

Ramesh Doddaiah: and all the top banks more or less all the banks also used to use that it it owns 50% of the market share. What it means is say for example if bank of America make transaction because you buy some New York stock exchange uh

Alberto Perdomo: Oh.

Ramesh Doddaiah: stocks. So it comes to us and uh for me the IO was the read and write was less than 50 microconds and

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Alberto Perdomo: Wow.

Ramesh Doddaiah: we can do the bandwidth of close to uh 350 to 400 GB.

Alberto Perdomo: Wow.

Ramesh Doddaiah: So it is exceptionally distributed storage with a huge bandwidth and amazing latency. So what it means is at any point in time we have close to 15 to 20 million read and write iOS. So I have to schedule 20 million IOS 20 million request coming from Bank of America or Oracle or all the databases. So managing that um I figured it figured it out that activity based storage will help a lot. The basic foundation behind that is called 8020 rule that you might already know.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: What 8020 rule is basically in 10 seconds is um the 20% of the say I will give example in the KV cache context. So the 20% of the KV caches uh say we put all the KV cache offloaded to the SE it might have say 100 terabytes of uh KV cache data SE file system but at any given point in time in that time window it's all time series based at any point in time say it is a 10 seconds time window or it can be 10 minutes or it can be 1 hour maximum.

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Ramesh Doddaiah: So we are trying to forecast what is going to be useful in the next 5 seconds, 10 seconds, 10 minutes or 1 hour so on. So based on that forecasting uh we pull the data from the disk and push it to the cache. So whenever uh in our case when uh if you're doing a benchmarking you do a lot of

Alberto Perdomo: Okay.

Ramesh Doddaiah: read veces does lot of read activities. So everything will be cached. cache in the sense uh it's a few terabytes of CPU RAMs just like what we have in the uh our clusters. So the difference is um assume right now we are getting a latency of I don't know what you were getting um in your chart assume the latency is milliseconds we can cut it down

Alberto Perdomo: Yeah.

Ramesh Doddaiah: thousand times because we are able to get in microsconds if it is in the RAM so that is exactly what it means activity

Alberto Perdomo: Yeah. Yeah. Absolutely.

Ramesh Doddaiah: based work means we forecast the activity which KV cache might be used may be needed sooner or ator and then we pull the data from the disk and put it to the CPU RAM.

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Ramesh Doddaiah: In this case, we can also push it to the GPU. You see,

Alberto Perdomo: Thank

Ramesh Doddaiah: so this is the simple way of explaining the activity based work.

Alberto Perdomo: you.

Ramesh Doddaiah: But when you say activity based, we have to keep a name for that. Uh activity based um um storage. Oh, abc or abs. I have to come up with a cool name for that. Um, I ABC looks

Alberto Perdomo: Um

Ramesh Doddaiah: cool.

Alberto Perdomo: it still looks cool activity base. Uh uh.

Ramesh Doddaiah: There's one more addition to this. Say for example,

Alberto Perdomo: Um

Ramesh Doddaiah: when we um offload the KV cache data to the disk, right now the SE file system is not doing probably any compression or encryption all those things. If we can compress the data on the disk,

Alberto Perdomo: yes.

Ramesh Doddaiah: we can save three times more or four times more or five times more. Right?

Alberto Perdomo: what's happening in some cases is that uh is not compressed but it's quantized

Ramesh Doddaiah: So

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Alberto Perdomo: uh in the GPU directly. So the KB block are already quantized.

Ramesh Doddaiah: yeah,

Alberto Perdomo: So they are

Ramesh Doddaiah: that's a the two different dimension.

Alberto Perdomo: smaller.

Ramesh Doddaiah: If you quantize that will also save the space.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: After quantizing you can still compress.

Alberto Perdomo: Yeah. Yeah. Exactly.

Ramesh Doddaiah: But the problem is the compression takes time.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: It'll add to the latency. So when you want to read the compress data from the disk because if it is a KV cache hit in the

Alberto Perdomo: Yeah.

Ramesh Doddaiah: disk, you have to decompress the data that will take lot of latency and then send it to the GPU. So in our idea, so assume we compress more or less everything that goes to the

Alberto Perdomo: Heat.

Ramesh Doddaiah: disk. But based on our uh time series analysis assume um the important 20% of the data that one we can u decompress it and keep it in the CPU RAM. So that way when there is a going to be KV cache 8 so data KV cache is already decompressed so we

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Alberto Perdomo: Yeah.

Ramesh Doddaiah: will not pay any addition latency one and also it is already in the CPU tire it's not in the disk. So that's a dual advantage for us.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: So this way even though we are able to compress it on the disk and keep a huge amount of KV cache but whatever the uh um the benchmarking needs benchmarking will never realize data is compressed and it will not realize the data is on the disk also it'll always read from the CPU that's the whole uh game so we are playing a game here based on the time series analysis so because of this if you have 100 terabytes of disk space also your latency and bandwidth will be exceptionally good that's the whole uh uh

Alberto Perdomo: Yeah.

Ramesh Doddaiah: solution so I don't know whether you you have used a time series uh um in your um earlier work or

Alberto Perdomo: Uh yeah.

Ramesh Doddaiah: not

Alberto Perdomo: Uh but do you mean time series models or time series databases or

Ramesh Doddaiah: um to build the time series algorithms like forecasting based algorithms right

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Alberto Perdomo: Okay.

Ramesh Doddaiah: so in this case we have to take the KV events that gets spilled out

Alberto Perdomo: Yeah.

Ramesh Doddaiah: from VLM all those things and based on that we have to build the time series algorithms or models so that models we Can obviously we have we have the GPUs we can run it on the uh slightly we can also use the some of the GPU resources but the point is uh the assume we have a time series model it continuously takes the data from the KV cache getting evicted or getting um um what is that what do you call um what is the right word for that KV LU right LU based events uh evictions and

Alberto Perdomo: Yeah. Yeah.

Ramesh Doddaiah: timing of the KV cache and the size of the KV cache it takes all these different uh uh variables and builds a single model.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: Do you understand the word um um what do you call um time series you know in a single time series it can take so many variables as input and it can it can as

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Alberto Perdomo: Yeah.

Ramesh Doddaiah: output it can um forecast one variable. In this case,

Alberto Perdomo: Yeah.

Ramesh Doddaiah: we just have to figure out which are the hot or hotest or the cold or coldest. So, we have different categories of um output that time series model gives. So, wherever it says hot, we will go we'll find such KV events and then pull it out of the disk and put it to the

Alberto Perdomo: Yeah.

Ramesh Doddaiah: CPU uh tire. So, we Yeah. Ask the questions. Yeah.

Alberto Perdomo: Yeah, I think this this make a lot of sense. And and actually uh um there is one thing uh in the let me check where it was in the right now there is um but we can use uh for an model to to do the forecasting of the time series if we can get them in the in tabular data. I'm not sure if that's if that's

Ramesh Doddaiah: Can can you can you come again please? I didn't I I u I could

Alberto Perdomo: Oh,

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Ramesh Doddaiah: hear Oh yeah yeah

Alberto Perdomo: you probably have heard the XG Boost model.

Ramesh Doddaiah: yeah ex boost model is fine. Uh we can use those things. So this is how it is, you know. Um say for example if you don't have lot of um resources to run the model you can go with a lightweight model also very lightweight classical uh model is also fine

Alberto Perdomo: Yeah.

Ramesh Doddaiah: the accuracy may not be that great but uh we'll do some oblation study to figure out which model works or you can also go with the deep learning models like LSTM CNN time series or transformer time

Alberto Perdomo: Yeah.

Ramesh Doddaiah: series models.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: So anyway we have the GPU so we can leverage it. We we don't have to like um use the GPU every second of the day right. So uh whenever we get the resources we can train that um time series models.

Alberto Perdomo: But your your idea is to kind of um build a model and uh that model gets trained and then it gets deployed uh and it's what makes the decisions.

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Ramesh Doddaiah: Um model yes model exactly.

Alberto Perdomo: I mean like you Yeah.

Ramesh Doddaiah: Yeah more or less. Yes, you're in the right direction. Yeah.

Alberto Perdomo: Because for example in the in LMD right now there is a a welled path that it's called uh predicted latency model and it uses an XG boost model

Ramesh Doddaiah: Mhm.

Alberto Perdomo: and it's trained on the fly because it's so lightweight. I mean it depends of course in the number of features and the and the latent space that that our problem requires but uh there are some cases where you want to to train the model on the fly because it learns the data or we can have it pre-trained and then uh kind of also trained in in real time with the data as as at the same time as you make predictions. Yeah, there are a lot of possibilities. But yeah, I agree that this uh this approach is very cool and it could be uh very

Ramesh Doddaiah: Yeah. So now that you said I'm not sure uh I was not sure about the XG boost if we let's use the

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Alberto Perdomo: novel.

Ramesh Doddaiah: existing infrastructure and existing solutions like what you said and build on top of that. If that doesn't give us very good accuracy, we can change the model uh later. But to start with,

Alberto Perdomo: Yeah.

Ramesh Doddaiah: we don't want to start everything from scratch. We will leverage all the existing uh infrastructure on LMD. Now the main point here is uh the time series it's called multi multivariate time series. So um we need to figure out which of which all these variables that we need to collect as part of the LRUS KB events that gets evicted or gets reused.

Alberto Perdomo: Yep.

Ramesh Doddaiah: So uh shall we go to the code? I need to figure out what all the events that are generated as of uh today by the LRU so that we can um

Alberto Perdomo: Basically you get uh two types of events when a block is uh is uh

Ramesh Doddaiah: um

Alberto Perdomo: how is it called? Uh let me check uh when well you get the block evicted uh event and the block uh stored I think it is.

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Alberto Perdomo: Um, let Wait.

Ramesh Doddaiah: evict evicted is good. Do we have information about how many times the KB u cash was

Alberto Perdomo: f***.

Ramesh Doddaiah: hit in the last say 5 seconds or 10 seconds. We need to know the frequency also.

Alberto Perdomo: Uh yeah. Um I don't

Ramesh Doddaiah: We can add we can add all those things.

Alberto Perdomo: think

Ramesh Doddaiah: One is we need to know the uh duration uh the first time when it when it was um used and the latest time when it was accessed so that we know whether it was accessed in the last 5 seconds or last 10 minutes ago. Example I'm

Alberto Perdomo: Ah but you yeah that that's also very cool.

Ramesh Doddaiah: giving

Alberto Perdomo: Um um there is this you probably already know this arc which is adaptative cache uh uh adaptative replacement cache. I think it's very it's a very old uh implementation adaptative replacement cache which basically instead of having a single alert uh you have a frequency based list and a recency based list and uh this is actually this was actually the first PR that I merged in VLM.

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Alberto Perdomo: Uh I did it for um for CPU

Ramesh Doddaiah: Okay.

Alberto Perdomo: offloading.

Ramesh Doddaiah: Uh okay. Uh do you want to share the code? You want to share the screen code?

Alberto Perdomo: Yeah,

Ramesh Doddaiah: Yeah.

Alberto Perdomo: let me share my screen. Okay, one second. Let me close this browser because I have I have I'm having issues with the with some software from from the from like from management in my laptop. Uh, and it is slow lately, but um, okay, let me go here and uh, PLM. Can you see my screen? Right.

Ramesh Doddaiah: Oh, yeah. It's coming up.

Alberto Perdomo: Okay. Yeah. Um, so let me go here

Ramesh Doddaiah: Yeah.

Alberto Perdomo: through closed. Uh, yeah. So this is the first PR that that I got merged in VLM. So I imple KV cash eviction policy. It was almost a year ago. And uh the code is very simple. Um the idea well I have an explanation here.

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Alberto Perdomo: Uh the core idea is that you have two lists. T1 is the res the recent cache which is uh the blocks that has been accessed one and T2 cache which is the list that

Ramesh Doddaiah: Mhm. Send

Alberto Perdomo: multiple times and then I have uh uh two ghost list that are B1 and B2 which basically track the recently uh recently evicted blocks. So in that case you have them in that list and you can bring them back faster.

Ramesh Doddaiah: Okay. Can uh about That can we say say for example we have 10 prefields

Alberto Perdomo: Yeah.

Ramesh Doddaiah: and say 20 decodes.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: Just for example if multiple decodes are accessing this do we figure can we figure out which decode accessed say uh recently and which decode accessed um uh first. Can we also capture the path?

Alberto Perdomo: Um I think so. Let me go to another thing. Uh because uh LMD

Ramesh Doddaiah: Yeah. I I'll get some water uh in a minute.

Alberto Perdomo: Okay.

Ramesh Doddaiah: Yeah.

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Alberto Perdomo: Okay. Cool.

Ramesh Doddaiah: Yeah.

Alberto Perdomo: So yeah,

Ramesh Doddaiah: Yeah.

Alberto Perdomo: basically I I'm I asked my agent to do a bit of research about the the events canonical form. As far as I recall, the canonical form is something like this. Uh sorry,

Ramesh Doddaiah: Mhm.

Alberto Perdomo: something like this. So you have uh um sorry the the event tag which can be whatever block stored or block evicted. Uh I don't you have um um the path this this code is from from a PR that I that I did

Ramesh Doddaiah: Okay.

Alberto Perdomo: for LLMD but this code I think it's it's already it's now present in VLM um they donated this connector. So I think in some for example the token ids or some other field could be used as a path because it's not used. Uh for example uh in the storage tier this part is it's not used. Uh this part is not used either. Uh, the block size is probably not used either. So in some cases maybe using the the secondary tiers which are uh either MVME or or or SE or whatever we can use one of these fields to to store the the file the yeah the the path to

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Ramesh Doddaiah: Okay. Okay.

Alberto Perdomo: the

Ramesh Doddaiah: So, uh in this case the block size right now is it not populated?

Alberto Perdomo: uh no uh well this is um yeah but this is an LMD. Let me see if uh my agency is uh uh well I'll come back to this. Um this all is in let's see try to find the code. Uh so this is VLM where is V1 V1? Oh, here V1 uh KB offload uh tiering factory manager. Let's see what block there is. I don't think there is uh okay so this is let's let me read a bit this is the tier of flowing manager multi-ter flowing orchestrator coordinates between a CPU primary tier with direct GPU access and zero or more secondary tiers storage network to provide hierarchical okay so yeah I don't think this is this is request state uh turn off load manager. Look up. Okay. Flash load. Touch. No. Override. Complete load in this. Okay. Let me see if I'm I'm trying to look for the the events canonical form that this is using.

00:32:10

Alberto Perdomo: Um, complete the store. Let me see if this found it. Okay, so yeah, in BLM. Okay, it's here. BLM config.

Ramesh Doddaiah: Okay,

Alberto Perdomo: Okay,

Ramesh Doddaiah: cool.

Alberto Perdomo: VLM. No worries. What? Okay, it's inside here. Uh config KV events. Okay, cool. Oh yeah, the KV events config. Well, let me double check. Okay, this is the configuration max Q size. Yeah, I don't think it's using the block size here.

Ramesh Doddaiah: and we can populate all those things because that will be useful down the line. Uh to build the time series one,

Alberto Perdomo: Yeah. Yeah. Definitely. Yeah. Block.

Ramesh Doddaiah: can you

Alberto Perdomo: Okay. Block stored. KB cache event. Okay. This is the base class and that we have block stored. Uh uh let me

Ramesh Doddaiah: Where's the time event access and frequency

Alberto Perdomo: see.

Ramesh Doddaiah: all those

Alberto Perdomo: Yeah, in this case I don't think it's it's that's recorded inside of the

00:33:23

Ramesh Doddaiah: things

Alberto Perdomo: block uh event. Okay,

Ramesh Doddaiah: M I see. Okay. This is event

Alberto Perdomo: these are yeah but this is for because the events are are sent in batches.

Ramesh Doddaiah: uh

Alberto Perdomo: Uh so this is the time of the batch. I don't think there is a time stamp in the in the block KV cash event. Yeah.

Ramesh Doddaiah: because uh one other thing is I see Maroon is that how we pronounce his name? Maroon.

Alberto Perdomo: Maru.

Ramesh Doddaiah: Yeah, Maroon.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: Um he keeps bringing up a lot of the session um um related uh uh ID right?

Alberto Perdomo: Yeah. The session. Yeah.

Ramesh Doddaiah: Yes.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: For us um in that case eventually we can also keep the session ID also. Uh that way it help us to prefetch a lot. The other thing is yeah this is one other thing which I wanted to ask currently VLM does it do the prefetch

Alberto Perdomo: um I think I thought it didn't uh because there are two cases there is one case when you are using just VLM and there is one case when you are using uh um how do you say the LLMD so I think from point 24 onwards VLM does the prefetch

00:35:01

Ramesh Doddaiah: Okay. Can it go the code? So are you saying that it can prefetch?

Alberto Perdomo: but

Ramesh Doddaiah: Uh when it prefetches, how does it figure out which is the forecasted KV blocks that is needed? I want to see that code. So

Alberto Perdomo: yeah that's because uh the The way blocks are hashed is the

Ramesh Doddaiah: forward.

Alberto Perdomo: based on a on a chain based on on the same parent. So it's the prefix uh hash. So you basically uh determine um like you go one block at a time to see how far you can go and have a hash that matches that whole prefix. But yeah, um uh it's

Ramesh Doddaiah: Oh wait, wait, wait, wait. You are only talking about prefix.

Alberto Perdomo: not

Ramesh Doddaiah: If the prefix matches then you're trying to pref prefetch.

Alberto Perdomo: Yeah. Yeah.

Ramesh Doddaiah: Uh okay. Okay.

Alberto Perdomo: Uh let me let me ask uh um you can tell me if BLM And Rhino reflections the K bit

Ramesh Doddaiah: So because

00:36:03

Alberto Perdomo: blocks.

Ramesh Doddaiah: for our time series activity based uh let me use the word ABC the for our ABC model um the foundation will be uh preetch. So prefetch is one other idea that we can also work.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: Preet is right now today.

Alberto Perdomo: Sorry,

Ramesh Doddaiah: So that is more I'm saying a preetch is

Alberto Perdomo: I lost you. Can you go again?

Ramesh Doddaiah: needed right now today in the uh field or in the market.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: So activity based idea is like umbrella. It's a huge idea in that uh we want to make prefetch work as

Alberto Perdomo: Yeah.

Ramesh Doddaiah: soon as possible. Um because say for example assume there's a prefix A and you said if there's a prefix matches uh if the longer is a prefix you'll try to prefetch that K blocks right but what say for example you go to the uh say I go to the uh restaurant I don't go alone you know I go always with my family so whenever we go it's three different peoples are going in a car and coming.

00:37:16

Ramesh Doddaiah: So if you know that if you are prefetching a prefix A, there's more chances that prefix prefix A, prefix B, prefix C are always accessed together.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: So this is across prefix prefetches,

Alberto Perdomo: Yeah.

Ramesh Doddaiah: right? So these are the simple things that we can make it work uh as of today uh step by step and

Alberto Perdomo: Yeah.

Ramesh Doddaiah: then you can build on top of that. But I still want to understand the prefetch code um in the

Alberto Perdomo: Yeah, let me see that. Uh, wait a bit. Uh,

Ramesh Doddaiah: VLM.

Alberto Perdomo: because you know I added to this uh this uh arch thing. Oh, I stopped sharing uh to this arch thing. I added um uh a few connectors and a few things. So, it can right now access uh uh the repositories and inspect the code. So I asked him to inspect the code and it's probably looking for it. So I mean it we we need to understand the code later but at least we can go

00:38:13

Ramesh Doddaiah: Okay.

Alberto Perdomo: directly to

Ramesh Doddaiah: Yeah. You know what? Um um give me one second.

Alberto Perdomo: it.

Ramesh Doddaiah: Uh I want to sh because this is a common problem which I see. I think it may be the Google Meet issue with the uh Mac. I I generally I So if you use zoom is is it the same

Alberto Perdomo: Oh yeah. Um I'm not sure if you No,

Ramesh Doddaiah: problem.

Alberto Perdomo: I mean this it's not a Google Chrome issue as far as I know. It is a Safari

Ramesh Doddaiah: Okay.

Alberto Perdomo: issue.

Ramesh Doddaiah: Okay. So next meeting we will try the zoom and um see if how it works.

Alberto Perdomo: Okay.

Ramesh Doddaiah: Yeah.

Alberto Perdomo: Okay. Cool.

Ramesh Doddaiah: And also importantly what I wanted to say is um uh for the next Friday's meeting. No, before that uh we will share some um um uh directions where we both can dig into this and come back with more homework so that we can go deeper

00:39:11

Alberto Perdomo: Yeah.

Ramesh Doddaiah: right. So preetch I'm going to take the prefetch uh I mean we both can take the prefetch at the same time that is

Alberto Perdomo: Yeah.

Ramesh Doddaiah: fine. So let's see let's make sure that we are working on the preetch. We want to understand the code and we want to understand what are the missing um um features that current prefetch has and try to see come up with the idea before we meet. So and when we meet we can go deeper and u ask cross

Alberto Perdomo: Yeah. Yeah. I agree. I agree on that.

Ramesh Doddaiah: questions.

Alberto Perdomo: Yeah. Yeah. Uh today it's the first, but yeah, I agree that we should do some prior work in order to to make the most out of this uh

Ramesh Doddaiah: Yeah.

Alberto Perdomo: hour.

Ramesh Doddaiah: Yeah. And uh it's fine if you don't have time but um having doing this homework will actually um make us more productive. We'll focus on this core idea later.

Alberto Perdomo: Yeah.

00:39:59

Alberto Perdomo: Yeah. Absolutely.

Ramesh Doddaiah: Yeah.

Alberto Perdomo: I completely agree. Yeah.

Ramesh Doddaiah: Did uh was there a action item on me to create a document um um for our this cool cool work

Alberto Perdomo: Um I I I created a sorry

Ramesh Doddaiah: or

Alberto Perdomo: um I created a document are getting notes but I will send it to you and we can because I haven't had uh anything yet.

Ramesh Doddaiah: That's fine.

Alberto Perdomo: Uh

Ramesh Doddaiah: At least if you can share me the link. No, at least I will keep adding more and

Alberto Perdomo: yeah. So uh I have just made you an editor and let me send you the link here.

Ramesh Doddaiah: more.

Alberto Perdomo: So yeah, we can have this share. I will link this uh this um document to our well I'll put it right now. Uh calendar I will take this one. This one. this one. Edit. Um, sorry. Okay, let's do it this way. You're about to make Okay. So, we have the link here now.

00:41:12

Alberto Perdomo: And if this has some something, it's taking a lot of time. But yeah, I definitely want to understand if if it's uh making the prefetch and based on what it's making the prefetch because I think it's been an active um you know in LMD one of the things that I did for this uh this offloading connector was actually I the events emission uh of the connector. So uh the the arctic uh the stored events and the evicted events uh I added them and the whole idea was to in the future be able to prefetch uh under some circumstances you know uh so yeah this this was in line with some of the work that I was doing upstream.

Ramesh Doddaiah: Okay. Uh, do you want to bring bring up that uh this one kash offloading Google docs.

Alberto Perdomo: Oh

Ramesh Doddaiah: Yeah.

Alberto Perdomo: yeah.

Ramesh Doddaiah: So I made few additional tabs. So the patents one uh added a few to I don't know why it's not refreshing for you. Yeah.

Alberto Perdomo: Okay.

Ramesh Doddaiah: And uh for the code part, right?

00:42:16

Ramesh Doddaiah: Uh at least for the um whatever you're discussing or um Okay, let me put it like this. Give me a second. um short-term

Alberto Perdomo: Yep.

Ramesh Doddaiah: um because uh what happens is when we discuss we come up with a cool um uh ideas all those things but eventually we'll forget I'm getting old I will definitely forget so keeping this um document with notes we all come up with a great ideas each and each each and every one of us will come up with great ideas in life but the problem is say if you have one cool idea did we ever go and file it in the redat as a patent did we really do the PC's no so we that's the problem so because of that having these um

Alberto Perdomo: Yeah.

Ramesh Doddaiah: notes we can always go back and see did we file it patent or did we finish the coding for that and all those things we can um uh that's why I made few tabs we just based on category feel free to add but uh the prefetch one.

00:43:22

Alberto Perdomo: Awesome.

Ramesh Doddaiah: Yeah. And the code part uh um so okay sorry I know I I cuted you when you were talking about the prefetch. So you said something that you implemented in the preetch right.

Alberto Perdomo: not something that I implemented but basically that the whole uh work that I one of the main things that I did

Ramesh Doddaiah: So

Alberto Perdomo: upstream was actually around the storage connector and the KV events around it and the whole goal was about the prefetch but I'm not sure the state of that right now since VLM implemented uh some part of it. So yeah, we will as long as we have some some pointers to start reading the code of the of the prefetching, we'll

Ramesh Doddaiah: Okay. So let me let me tell you what I'm thinking about.

Alberto Perdomo: see.

Ramesh Doddaiah: Say for example uh these are the ideas. Say the prefill computes the KV cache and we can move

Alberto Perdomo: Yeah.

Ramesh Doddaiah: the KV cache. Uh that's that's about um what is what is her name guy right?

00:44:24

Ramesh Doddaiah: Neil guy or Neil guy I don't know what is her name guy right so

Alberto Perdomo: Oh yeah.

Ramesh Doddaiah: she was trying to do the P2P that is moving the data from prefill to the decode or one node to the other node but the problem

Alberto Perdomo: Yeah.

Ramesh Doddaiah: is if you do that you have to go through the fabric say your PCI assume your PCI

Alberto Perdomo: Yeah.

Ramesh Doddaiah: bandwidth is example I'm just giving example assume your PCI bandwidth is a 16 Gbps but your fabric is only 8 Gbps. That means even though you have a lot of resources using the PCA bandwidth,

Alberto Perdomo: Yeah.

Ramesh Doddaiah: you are not using it. You are only using the 50% um uh lower fabric bandwidth to move the data from one uh pier to the other pier.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: And my question is uh what if you move the data from the pre-fill node to the disk say the parallel file system

Alberto Perdomo: Yeah, that would that would be Yeah,

Ramesh Doddaiah: that way.

Alberto Perdomo: because then the

Ramesh Doddaiah: Yeah.

00:45:30

Ramesh Doddaiah: Keep asking. Yeah. Yeah. Yeah. So that way so uh assume uh there are multiple decode nodes

Alberto Perdomo: right

Ramesh Doddaiah: which which it wants the u kach just got um dstaged by the prefill node to the se. So we can move transfer the kv cache from the se into all these multiple decode nodes before it actually needs it. So we are trying to do distributed preetch here. So distributed prefetch we are able to do even in the VLM.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: VLM is a single node. No it's not like LMD but with our model we can figure out all the different path through which the KV cache is needed and try to keep the KV caches as local as possible for that particular node. That way when it accesses you get it have very short um low latency for those things. I will make a note of those things. These are all cool ideas. Um, prefill and decode thing move. Let me uh prefill the computes new KV caches and uh these stages into parallel system.

00:47:03

Ramesh Doddaiah: Multiple multiple decodes will our preach all in decode. Remove the code to make this cash access local bandwidth instead consume PCI. So basically what we're trying to do is we're trying to optimize right we have um PCI path uh we have a fabric path that we have several PCI path. So we need to make sure all these different parts are uh utilized 100% of the time. If not we'll get a uh skewed.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: No only if fabric is getting used and PCI is not getting used we have a resources but we're not doing efficientuling um we'll waste the bandwidth.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: Okay cool.

Alberto Perdomo: Absolutely.

Ramesh Doddaiah: So I think this is what um this is what exactly happens when we discuss we come with a more ideas and more problems and a more uh valuable contribution we can make. So I think I I like that what we are doing. Uh we we'll keep this going forward this Friday's meeting. I don't know if it works for you at this slot every

00:48:42

Alberto Perdomo: Definitely.

Ramesh Doddaiah: week.

Alberto Perdomo: And and we can I mean we can take uh even more time. So that's fine for me.

Ramesh Doddaiah: Okay. Okay. Cool.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: So I come from um 8 to 9 morning I do have a research meeting.

Alberto Perdomo: Awesome.

Ramesh Doddaiah: So after the research meeting I will uh join you always at 9:00 maybe a few minutes late here and there but um if we want to extend the meeting then definitely uh we'll figure out how this goes next couple of weeks. I have one more meeting at 10:00.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: I will try to postpone that to 10:30 so that um if you want we can extend it going forward our meeting.

Alberto Perdomo: Awesome. Awesome. That's great.

Ramesh Doddaiah: Yeah, this actually more interesting you know uh we discussed all those things. It's just that what I'm missing right now is I do not have um insight into the VLM and LMD code to understand what exactly has been done as of today. So I have to spend some uh time and put some bandwidth into that to understand this prefetch and all these KV events.

00:49:41

Ramesh Doddaiah: Uh thankfully you showed few things for me but I have to go deeper to see the uh

Alberto Perdomo: Yeah. So basically uh what happens uh I'm sure you're pretty familiar with with most of this but when you when you

Ramesh Doddaiah: constraints.

Alberto Perdomo: are when you do the prefill you basically uh compute all the KV values for the for to then compute mention of the sequence length and then decode um is a is a process that's usually uh known as to be auto reggressive. So it's sequential. So in the decode you basically are reading and potentially in some cases extending the KB cache uh because you are using the the KB cache values to compute or to predict the the next token and it's one token at a time. So that's why why it is auto reggressive and that's why is sequential and that's why uh uh decode in some cases can be can be slow. So it depends.

Ramesh Doddaiah: Yeah. How about about this Alberto?

Alberto Perdomo: I mean Yeah. Go ahead.

Ramesh Doddaiah: Um say for example you are um a customer you are a rich customer and you pay lot of tons of uh money to get um they say cloud But uh um uh session.

00:51:02

Ramesh Doddaiah: Okay, assume you are the diamond customer. You pay tons of money. Assume I I'm the bronze customer.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: We are still using the same cloud, but I'm a bronze customer. You are the diamond customer. So, you have more weightage. So, we both have to do the decoding.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: Now, for the priority, you know, if uh I'm doing the decode, you will not be able to get the more resources. Though I'm the bronze customer, uh I'll be occupying that um decode node.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: Let me uh change it this way. So you said it's auto reggressive decode because it's a more sequential.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: What what if we could do multiple trajectories that we are trying to do for you as a

Alberto Perdomo: Yeah.

Ramesh Doddaiah: diamond customer? The decode is decode is trying to do uh multiple multiple my daughter I think she just woke up. She's making sounds. Yeah. So what I was trying to say is um maybe you have more insight here.

00:52:06

Ramesh Doddaiah: So the decode is not a single node. It can be any number of decodes.

Alberto Perdomo: Um yeah so sorry there is one thing that you probably know already too

Ramesh Doddaiah: So

Alberto Perdomo: which is called a speculative decoding.

Ramesh Doddaiah: Mhm.

Alberto Perdomo: which basically speculative decoding uh what it does is uh it has a smaller model load in the in the GPU and with that small model you propose a few draft tokens. So it's like saying okay um you have this question the answer can start with these five tokens five or not five five sequences and then the bigger model chooses the best one.

Ramesh Doddaiah: Okay,

Alberto Perdomo: So it kind of gets a a kind of a head start. So that happens

Ramesh Doddaiah: got it. Agreed.

Alberto Perdomo: that

Ramesh Doddaiah: But assume we do that now. Okay. And uh tomorrow tomorrow you come back and you again you start your prompts.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: So the learning that you have on Friday will be forgotten by the uh speculative decoding on

Alberto Perdomo: Yeah.

00:53:13

Ramesh Doddaiah: Saturday.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: So that is where we need to save that learning. So speculative decoding as you said if it's five different trajectories it is doing and it has found one trajectory to be

Alberto Perdomo: Yeah.

Ramesh Doddaiah: good. This is like a reinforcement learning. that the policy is learning what is the right direction for Alberto uh the speculative decoding did because you are you are a diamond you have a diamond session unlike me brown session so all the learning that happens inside this we don't want to forget it we want to put it into the disk so

Alberto Perdomo: Yeah.

Ramesh Doddaiah: that we build a policy specifically for say diamond customer the reason is instead of having five different speculative decoding assume you always use

Alberto Perdomo: All right.

Ramesh Doddaiah: Only three or two. So we know that for you three is good enough and for me maybe example I'm just giving

Alberto Perdomo: Yeah.

Ramesh Doddaiah: example maybe seven is good enough that way that policy right now um it's not learned per session or per user it is just a speculative decoding one configuration we will say that okay uh five different trajectories will take the same thing works for a bronze customer or for the this one diamond customer.

00:54:25

Alberto Perdomo: I see him.

Ramesh Doddaiah: So basically what we're trying to do is if you are a diamond customer I'm trying to give you BMW. If not for me as a bronze customer I will try to use a Toyota Corolla. Yes. End of the day your resources.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: Okay. Again uh this because you brought up all these pro uh points and I'm um thinking deeper and uh deeper. Um I'll take a step back. I will I'll make a note of that. Uh uh one second.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: Um

Alberto Perdomo: But yeah definitely there are a couple of things that because um in some I haven't done much uh PD disagregation work. Yeah, I need to to think about it about how um how the

Ramesh Doddaiah: Mhm.

Alberto Perdomo: decode pot uh how n decode codes can use the the KB cache at the same time or even if they do I'm not sure if they do because since it's a sequential process I need to understand better um

00:55:28

Ramesh Doddaiah: Yeah, my goal my goal for this uh our session is on Friday session is I want to understand each and every uh layer of the LMD and VLMs and LM cash and moon cake. I want to get into like um not expertise on one particular path or one particular uh module in the inference. I want to know the complete I want to be like architect for this.

Alberto Perdomo: Yeah. Yeah.

Ramesh Doddaiah: I I can easily understand all the KV cache offloading all the in and out.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: I can uh easily relate and I can give design solutions architecting I can architect the solutions but for the inferencing and the PD disagregation what you said and the emerging agentic workload for all those things I want to also learn a lot. So this actually gives me opportunity to uh openly ask all the questions and go in any direction which we want and learn whatever we are interested in. There's no restrictions here.

Alberto Perdomo: Yeah, absolutely. That makes a lot of sense.

Ramesh Doddaiah: Yes. Yes.

00:56:26

Alberto Perdomo: I like this approach.

Ramesh Doddaiah: Yeah. So we are willing to do the hard work so we can learn whatever we like.

Alberto Perdomo: Yeah,

Ramesh Doddaiah: Oh that will be cool. And oh uh one last question you know I know we have only three more minutes.

Alberto Perdomo: exactly.

Ramesh Doddaiah: Um um I have one-on-one with um uh Mishi at uh 2 or 2:30 on Fridays.

Alberto Perdomo: Okay, cool.

Ramesh Doddaiah: I actually wanted to ask him uh what are the best self? Um, you know, I pasted some of the um Dell links. So those were some of my ideas.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: Um I was um reading some of those um initiatives. So um they wrote the blog after I joined Redat the G3.5. So that is one of the cool ideas hackathon which I had won.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: My point is um with Mishi I need to ask because the Delhi is saying that they're able to get a few hundreds of GBs but SE we don't know what is the limit right so I'm going to bring up in the Mishi's meeting today and share share with you all the

00:57:29

Alberto Perdomo: Yeah.

Ramesh Doddaiah: discussion that we have but um do you think uh the chart that you write now which you have if you can put it to the um slack

Alberto Perdomo: Give me one second. I can give you um okay cash of loading quen this is the five way right yeah so let me export it as a PDF and I will send it to you let's see if if it is if it's uh cool because I I added recently this uh this um PDF yeah I think here you will have all the all the tables and most important ly. Well, yeah, there are some that are missing data, but yeah, you have here all the all the data from the report this.

Ramesh Doddaiah: most of it.

Alberto Perdomo: So,

Ramesh Doddaiah: Yeah.

Alberto Perdomo: to

Ramesh Doddaiah: Okay. I will mention this to in the slack saying that after we went what is the networking

Alberto Perdomo: you,

Ramesh Doddaiah: changes you did now it is not a pod networking so use RDM is

Alberto Perdomo: let me uh Yep.

Ramesh Doddaiah: it

00:58:41

Alberto Perdomo: It's basically using the the fast nicks uh uh with PCI pass through uh it's doing well I have yeah actually let me

Ramesh Doddaiah: okay

Alberto Perdomo: check because I think

Ramesh Doddaiah: why don't So because um um you did all this work why don't you put it in the slack so that I can ask follow-up questions to machine in the slack itself.

Alberto Perdomo: okay

Ramesh Doddaiah: Uh so I want them to know that you did all this work and you got amazing number. This is the beginning of the se good numbers. Now we need to build on top of that we need more se.

Alberto Perdomo: Yeah, exactly. So basically,

Ramesh Doddaiah: Yeah.

Alberto Perdomo: yeah,

Ramesh Doddaiah: Okay.

Alberto Perdomo: I will add to the

Ramesh Doddaiah: Yeah. Okay. Cool. Uh please do that uh so that I can talk to him in the meeting.

Alberto Perdomo: thread.

Ramesh Doddaiah: And the last thing is um I know I have to jump for other meeting. Last thing is um I will um start reading the code in the VLM and LMD the existing prefetch and

00:59:35

Alberto Perdomo: Yeah. you.

Ramesh Doddaiah: um

Alberto Perdomo: I will point you right now when the agent finishes where where is the code exactly so we can dive directly into

Ramesh Doddaiah: okay I put action item tab.

Alberto Perdomo: it.

Ramesh Doddaiah: So for me uh I will say that uh go deeper in the in the Kash offering document. go deeper into preetch um logic. check in VLM and LMD. So this is what I will be doing and exchanging the notes with you for you. Uh I I think I will I will leave it to you.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: So uh you can uh uh put it whatever you

Alberto Perdomo: Yeah. Do I will probably start doing the same to understand this and then when we have more knowledge about

Ramesh Doddaiah: like

Alberto Perdomo: this we can just uh I'll be deciding.

Ramesh Doddaiah: sync up.

Alberto Perdomo: Yeah.

Ramesh Doddaiah: Yeah. And uh Fridays is officially we're going to meet for this work,

Alberto Perdomo: Yeah.

Ramesh Doddaiah: right?

Alberto Perdomo: Yeah.

Ramesh Doddaiah: But uh if something interesting comes we can always meet earlier

Alberto Perdomo: Yeah. Oh, probably uh most of the weekends I'm able to meet,

Ramesh Doddaiah: also.

Alberto Perdomo: but it depends because my girlfriend has rot rotative shifts. So, so when when is her free weekend, I'm not allowed to work, but the other ones I'm allowed to work.

Ramesh Doddaiah: Got it.

Alberto Perdomo: So,

Ramesh Doddaiah: Yes. Yeah. Got it.

Alberto Perdomo: yeah.

Ramesh Doddaiah: Yeah. We will do that. Huh. I have to jump for other meeting but it's nice meeting you.

Alberto Perdomo: Okay.

Ramesh Doddaiah: Thanks man. Bye.

Alberto Perdomo: All right.

Transcription ended after 01:00:57